

BOTNET DETECTION FROM NETWORK TRAFFIC

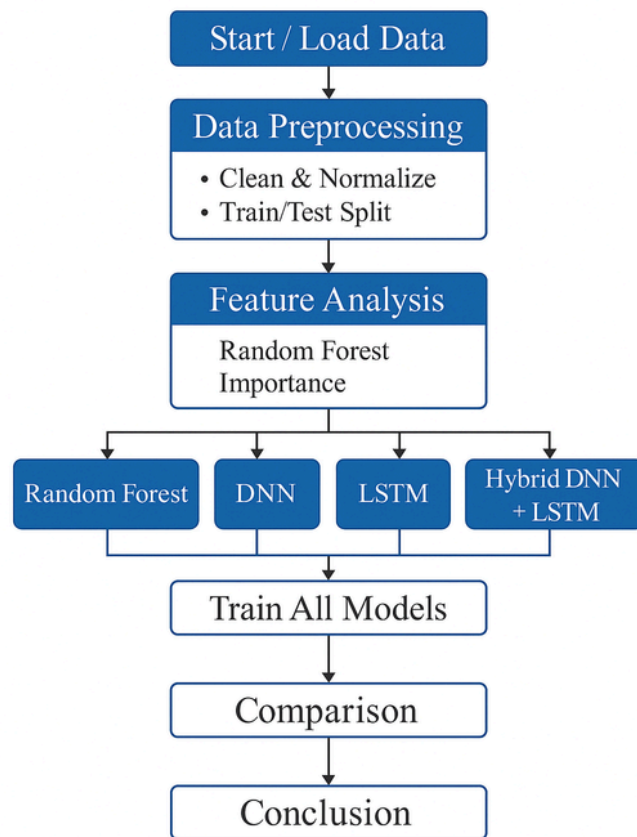
[A Machine Learning & Deep Learning Hybrid Approach]

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INTRODUCTION

Botnets disguise their malicious traffic to look like normal network activity, making detection challenging. This project analyzes NetFlow features and evaluates **Random Forest**, **DNN**, **LSTM**, and a **Hybrid DNN-LSTM** model to classify traffic as Normal or Botnet. Our goal is to identify which model provides the most accurate and reliable detection.

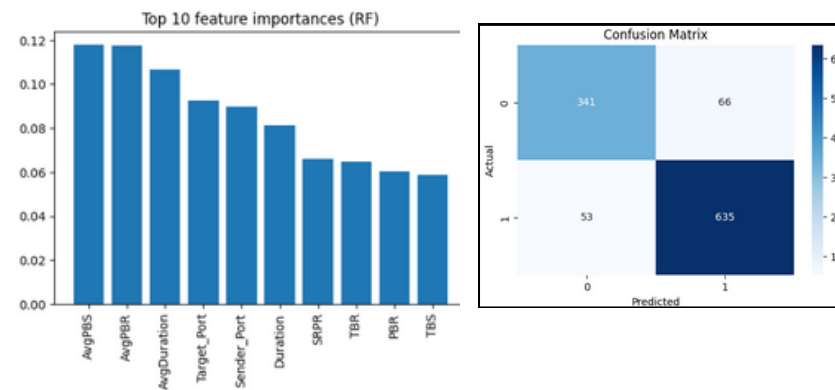
METHODOLOGY



DATASET

- 5,472 network flows with 19 NetFlow features.
- Includes duration, packet counts, byte rates, payload size.
- To prepare the data, non-essential identifiers such as IP addresses and port numbers were removed, missing values were handled, and all numerical fields were normalized using StandardScaler.
- The dataset was then split into 80% training and 20% testing.

EXPERIMENTATION



Random Forest Model

This served as our baseline classifier, evaluated using 5-fold stratified cross-validation to ensure balanced performance. We tuned key hyperparameters to optimize the F1-score, improving its detection ability.

Deep Neural Network (DNN) Model

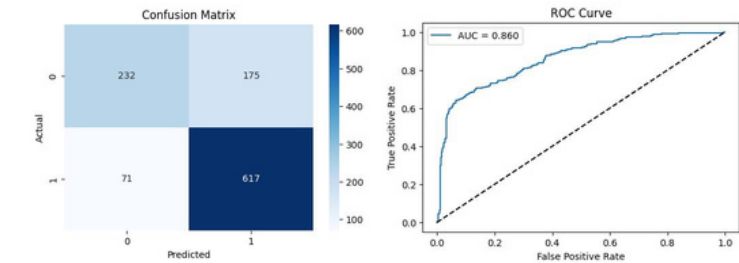
The DNN model is designed to learn feature-level patterns from network flow data.

Long Short-Term Memory (LSTM) Model

The LSTM model focuses on capturing temporal and sequential patterns in network traffic.

HYBRID DNN-LSTM MODEL

By combining the DNN and LSTM outputs, the hybrid model leverages both feature interactions and sequential dependencies, achieving superior accuracy.



RESULT

Model	30 ep	50 ep	100 ep
LSTM	81.55%	83.38%	82.47%
DNN	82.92%	82.59%	83.84%
Hybrid	77.63%	79%	83.29%

Model Comparison

- Random Forest Accuracy: 89%
- The DNN model achieved the highest accuracy, peaking at 83.84% at 100 epochs.
- LSTM also performed strongly, reaching 83.38% at 50 epochs.
- The Hybrid model improved with more training, matching competitive accuracy at 100 epochs.
- Overall, all models exceeded 80%, confirming deep learning as effective for botnet detection.

CONCLUSION

All models performed well above 80%, with DNN achieving the best deep-learning accuracy and LSTM close behind. The Hybrid model also showed strong results and we think its the better fit because it captures both feature and temporal patterns, making it more reliable for real, changing network traffic. Random Forest scored highest here since the dataset is static, but the Hybrid model is better suited for dynamic, evolving traffic.